

Datathon'25

Team ACE Engineers

Team Members:

Shahitty Md Yash Siddque

Alam Mir Tazwar



Problem Statement

Finance teams need timely visibility into weekly cash movements to manage short-term liquidity risks, understand volatility drivers, and plan sustainably for the medium term.

However, transaction-level cash data is noisy, making it difficult to:

- Identify true weekly cash patterns
- Anticipate near-term liquidity pressure
- Detect unusual cash movements early

Objective

- Analyse weekly cash inflow and outflow
- Forecast short-term (1-month) and medium-term (6-month) cash flow
- Identify anomalies and key risk drivers
- Provide a practical dashboard for finance decision-making

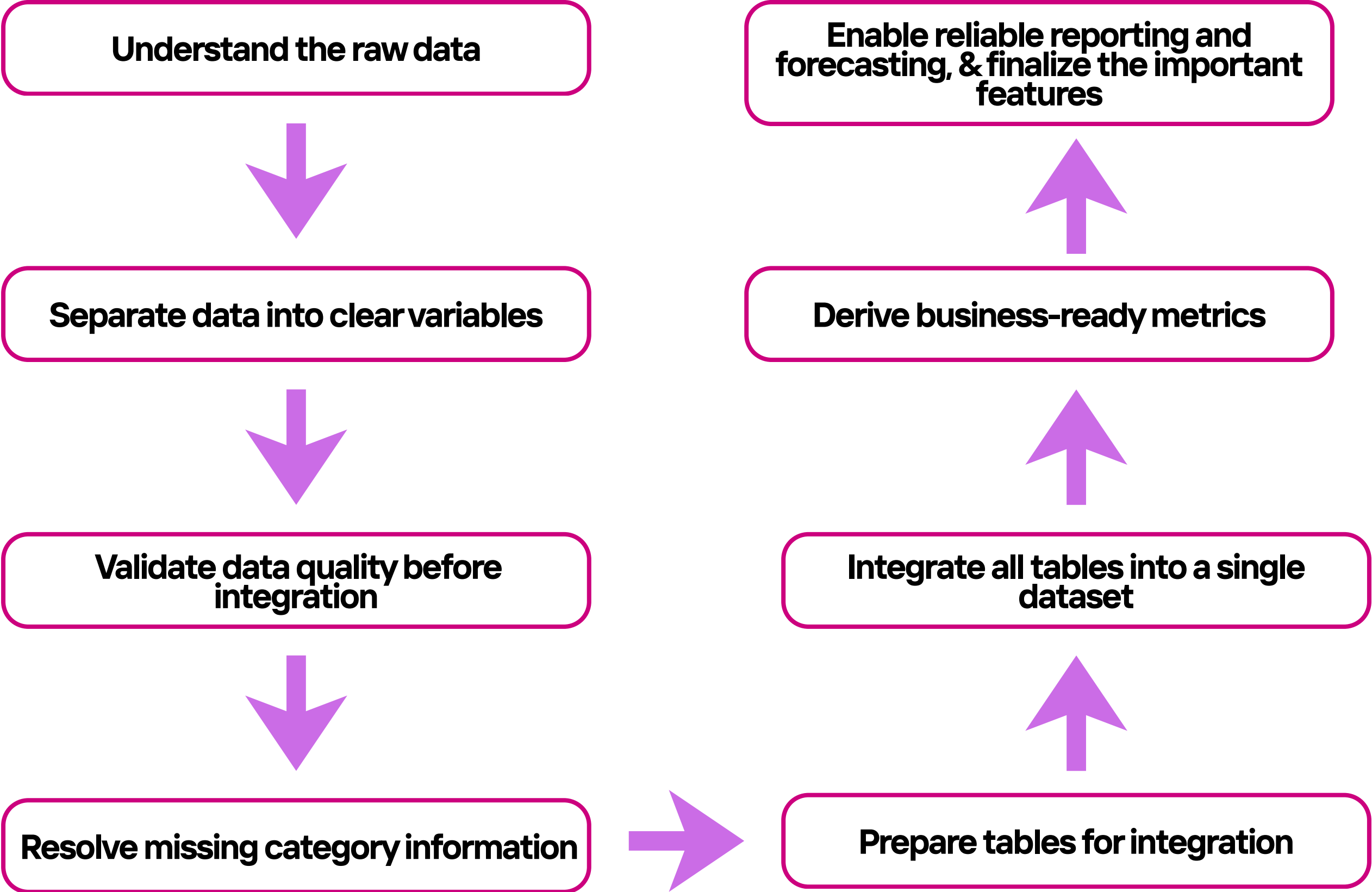


References:

Dashboard: <https://lookerstudio.google.com/reporting/5e1b0885-9687-4449-a345-3e0a5fc9f43f>

Codes: <https://colab.research.google.com/drive/1UGUIp3Cik1VuhX8FQUwK8svtgd0Vq9fd?usp=sharing>

Data Preparation & Integration Approach for Cash Flow Analysis



```
file_path = "/content/Datathon Dataset.xlsx"
data_main = pd.read_excel(file_path, sheet_name="Data - Main")
cash_balance = pd.read_excel(file_path, sheet_name="Data - Cash Balance")
category_link = pd.read_excel(file_path, sheet_name="Others - Category Linkage")
country_map = pd.read_excel(file_path, sheet_name="Others - Country Mapping")

print(data_main.shape)
print(cash_balance.shape)
print(category_link.shape)
print(country_map.shape)

(84528, 28)
(8, 4)
(25, 4)
(8, 3)

category_link = category_link.rename(columns={
    "Category": "Flow Type" # or "Cash Flow Type"
})

# Create mapping: Category Name -> ID
category_to_index = dict(
    zip(category_link["Category Names"], category_link["ID"])
)

data_main["Category Index"] = data_main["Category Index"].fillna(
    data_main["Category"].map(category_to_index)
)
```

```
# Create mapping: Category Name -> ID
category_to_index = dict(
    zip(category_link["Category Names"], category_link["ID"])
)

data_main["Category Index"] = data_main["Category Index"].fillna(
    data_main["Category"].map(category_to_index)
)

data_main["Category"].isna().sum()
np.int64(0)

merged = data_main.merge(
    category_link,
    left_on="Category Index",
    right_on="ID",
    how="left"
)

merged[["Category", "Category Names"]].drop_duplicates().head(15)
```

```
# Standardize 'Category Names' in category_link (remove hyphens, ensure single space, lowercase)
category_link["Category Names"] = category_link["Category Names"].str.replace("-", " ", regex=True).str.lower()

# Recreate the mapping with standardized names
category_to_index = dict(
    zip(category_link["Category Names"], category_link["ID"])
)

# Standardize 'Category' in data_main to match (remove hyphens, ensure single space, lowercase)
data_main["Category"] = data_main["Category"].str.replace("-", " ", regex=True).str.lower()

# Rebuild the mapping to fill 'Category Index' in data_main
data_main["Category Index"] = data_main["Category Names"].map(category_to_index).fillna(
    data_main["Category"]
)

# Re-merge the data with the cleaned columns
merged = data_main.merge(
    category_link,
    left_on="Category Index",
    right_on="ID",
    how="left"
)

# Now, let's check the unique values again after cleaning and re-merging
display(merged[["Category", "Category Names"]].drop_duplicates().head(10))
```

Category	Category Names
0	no
48218	payroll

Weekly Cash Flow Dynamics and Net Liquidity Position

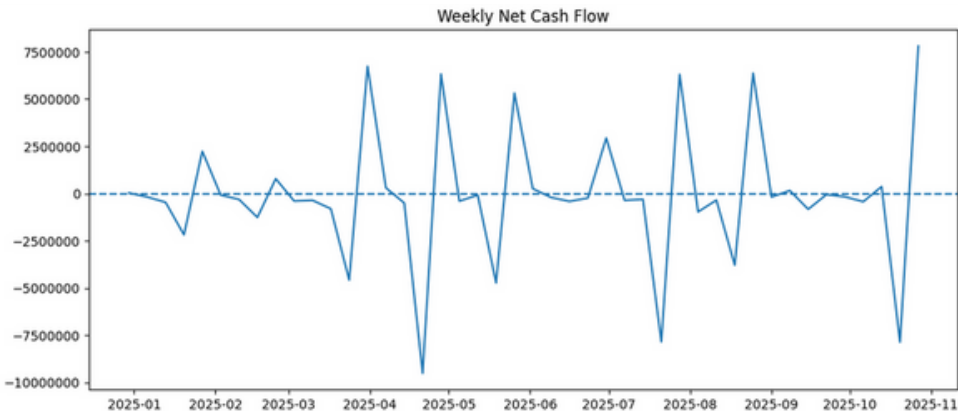


Chart : Weekly Net Cash Flow

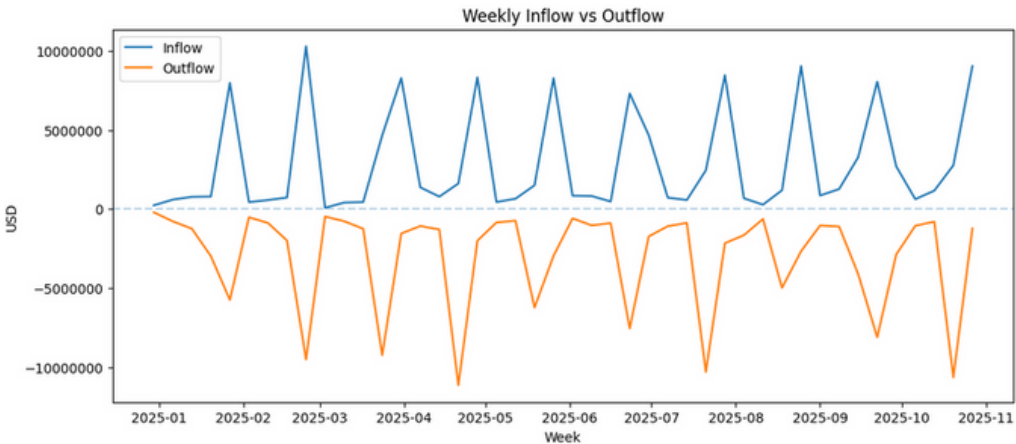


Chart : Weekly Net Cash Flow

	Week	Weekly Inflow	Weekly Outflow	Net Cash
0	2024-12-30	234,669.38	-203,175.12	31,494.26
1	2025-01-06	595,400.05	-776,322.43	-180,922.38
2	2025-01-13	772,735.51	-1,237,352.50	-464,616.99
3	2025-01-20	790,479.55	-2,974,405.20	-2,183,925.65
4	2025-01-27	7,976,897.65	-5,741,963.26	2,234,934.38
5	2025-02-03	439,185.66	-517,229.89	-78,044.23
6	2025-02-10	568,516.30	-879,528.30	-311,012.00
7	2025-02-17	729,198.64	-1,990,734.25	-1,261,535.60
8	2025-02-24	10,287,893.98	-9,505,199.37	782,694.61
9	2025-03-03	78,930.06	-473,794.83	-394,864.77
10	2025-03-10	409,505.53	-767,680.83	-358,175.30
11	2025-03-17	438,368.49	-1,243,598.77	-805,230.28
12	2025-03-24	4,654,785.21	-9,230,641.14	-4,575,855.93
13	2025-03-31	8,280,186.18	-1,546,377.43	6,733,808.74
14	2025-04-07	1,373,707.16	-1,073,162.40	300,544.76
15	2025-04-14	786,417.90	-1,288,864.00	-502,446.10
16	2025-04-21	1,615,386.03	-11,125,541.24	-9,510,155.21

Key Findings:

- Strong recurring seasonality.
- Net cash fluctuates around zero.
- Large negative spikes are driven by outflows.
- Positive spikes follow shortly after negative weeks.
- Volatility is operational, not random

! Weekly cash volatility is primarily driven by the timing of large outflows rather than instability in inflows, making cash timing the key short-term liquidity risk

✨ We prepared the data and analysed weekly inflows and outflows to reflect how cash is managed in practice. The pattern shows regular, predictable cycles driven by billing and payment schedules. Large scheduled payments occasionally push net cash negative, but these are typically followed by strong inflows. This indicates that short-term cash risk is mainly a matter of timing, not insufficient income.

• Table: Weekly Cash Flow

Weekly Net Cash Volatility & Anomaly Identification

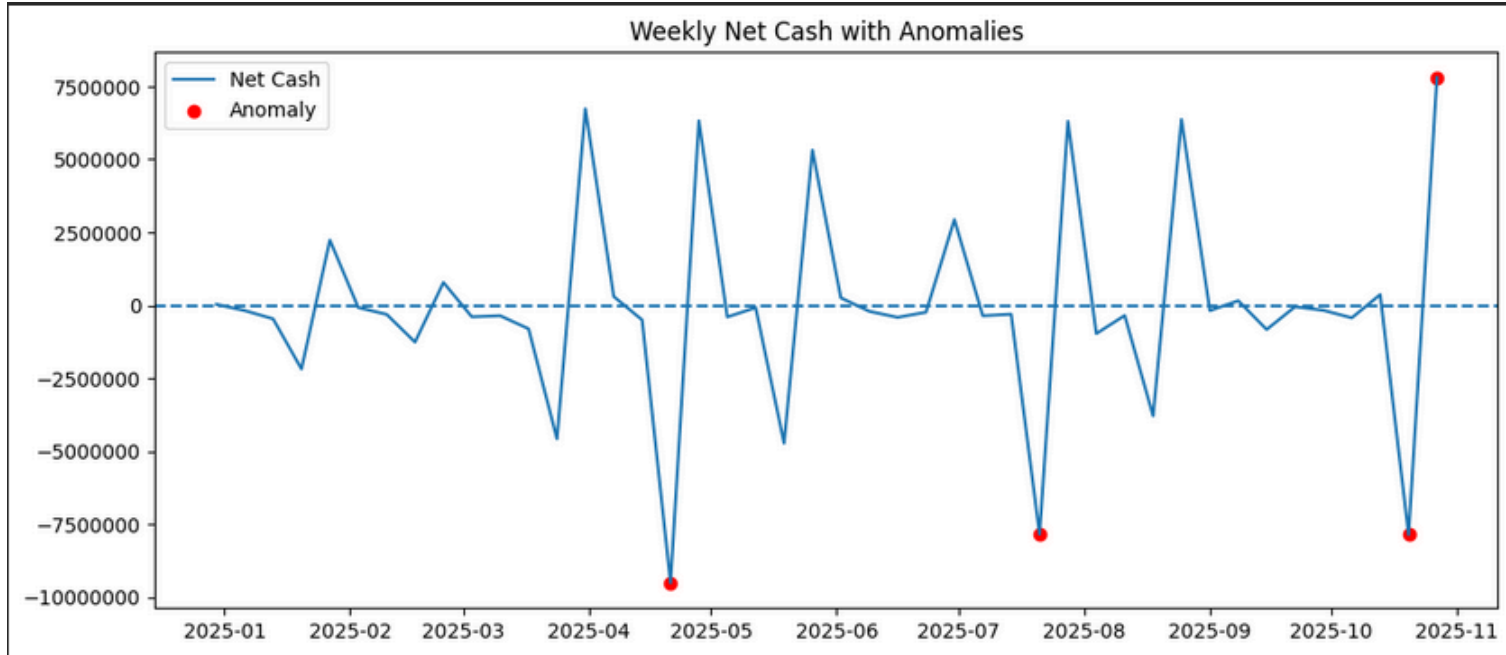


Chart: Weekly Net Cash Flow with Anomaly Flags



Anomalous Weeks: Inflow vs Outflow Breakdown

Key Insights:

- Weekly net cash follows a strong seasonal pattern driven by regular billing and payment cycles.
- A small number of weeks show extreme net cash movements due to unusually large or clustered inflows or outflows.



Anomaly detection helps finance teams focus attention on weeks with unusual cash movements caused by operational timing, not underlying business instability.



Weekly net cash generally follows a predictable cycle, reflecting routine business operations. However, certain weeks experience unusually large inflows or outflows that push cash far from its normal range. These are not data errors, but operational events such as bulk payments or delayed collections. By flagging these weeks as anomalies, finance teams can quickly review and assess potential short-term liquidity risks.

Weekly Cash Flow Forecast Using Seasonal Time Series Model

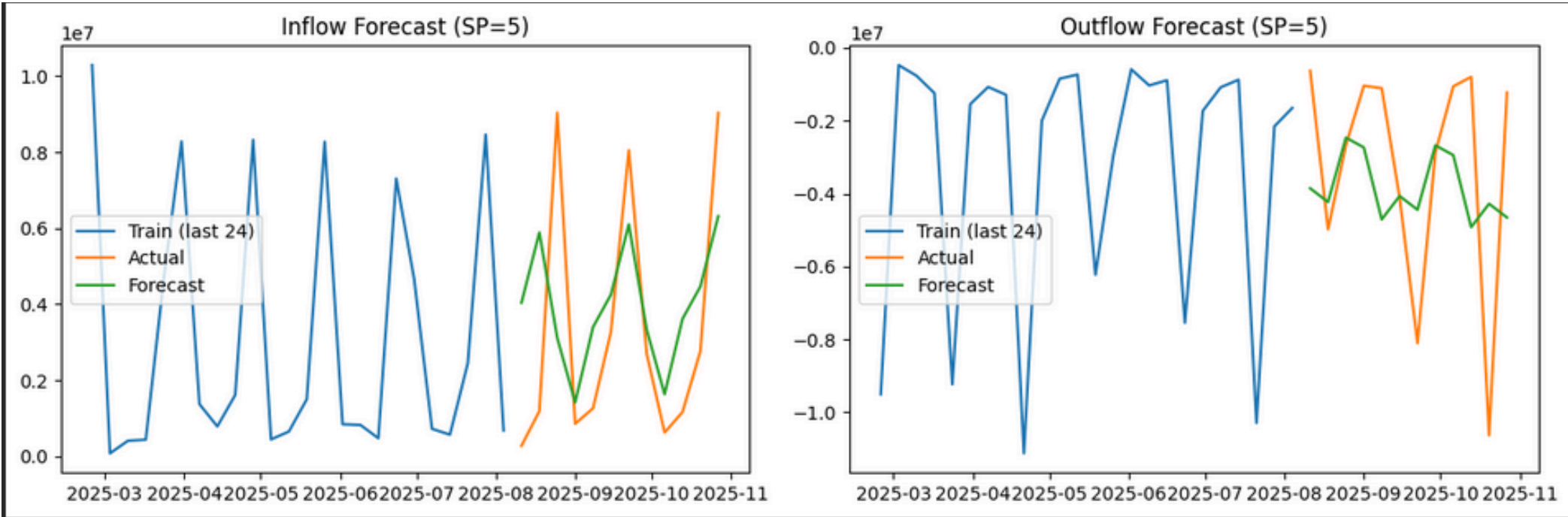


Chart: Weekly Inflow & Outflow Forecast Comparison

```
SP=5
Inflow  → MAE=2,376,258.01, RMSE=2,866,370.76
Outflow → MAE=2,429,447.75, RMSE=3,067,632.14
```

MAE & RMSE (Input And Output Flow)

Step1

To evaluate forecast performance objectively, the data was split into:

- Training set: historical weeks excluding the most recent 12 weeks
- Testing set: the most recent 12 weeks

This approach simulates a real-world scenario where future values are unknown.

Step4

Forecasts were generated for the test period and compared against actual values.

Performance was evaluated using:

- MAE (Mean Absolute Error)
- RMSE (Root Mean Squared Error)

The model with seasonal period = 5 achieved the lowest RMSE for both inflow and outflow, indicating better handling of large forecast errors.

Step2

Multiple forecasting configurations were tested.

Holt–Winters Exponential Smoothing was selected due to:

- Clear seasonal patterns in the data
- Need for an interpretable model for finance stakeholders

Seasonal periods of 4 and 5 weeks were evaluated.

Step3

The model was trained separately on:

- Weekly Inflow
- Weekly Outflow

Additive trend and additive seasonality were used to capture recurring billing and payment cycles.

Key Findings

- Weekly cash movements exhibit strong and predictable seasonality.
- Forecast accuracy is sufficient for near-term liquidity planning.
- The model provides finance teams with early visibility into upcoming cash surplus or deficit periods

One-Month Weekly Cash Flow Forecast & Liquidity Outlook

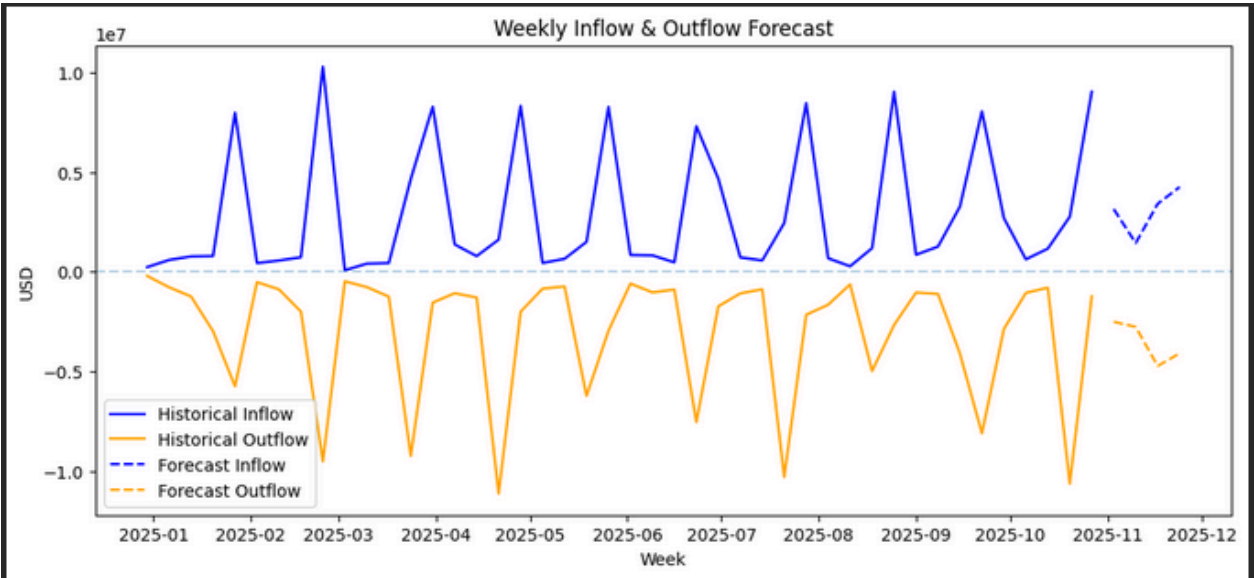


Chart: Weekly Inflow & Outflow Forecast

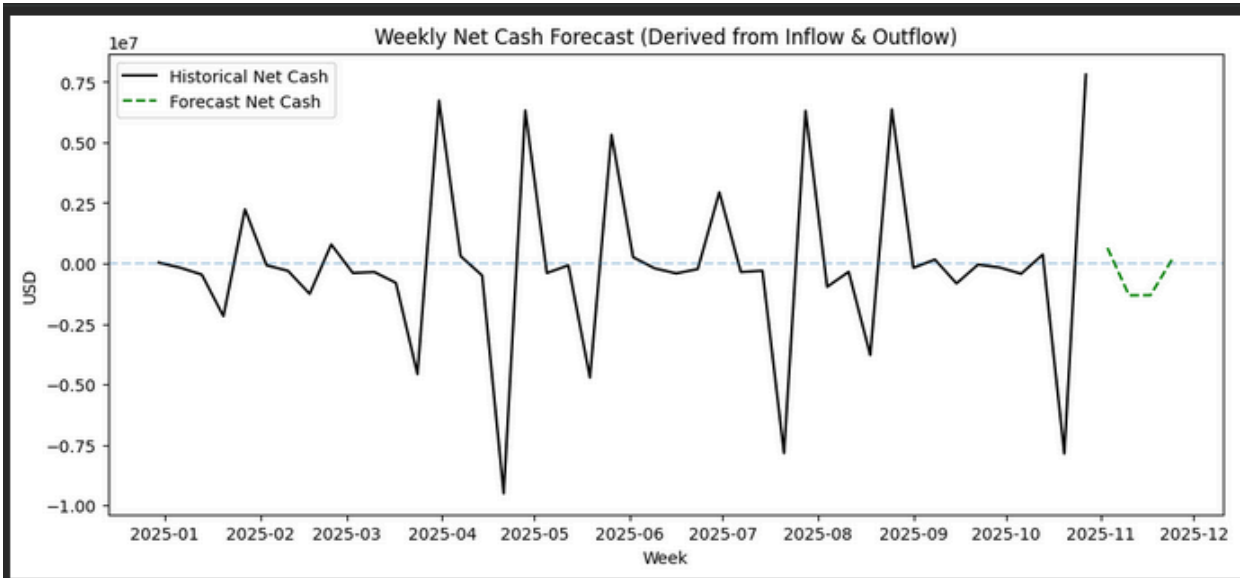


Chart: Weekly Net Cash Forecast

```
model_in = ExponentialSmoothing(  
    ts_in, trend="add", seasonal="add", seasonal_periods=5  
)  
.fit()  
  
model_out = ExponentialSmoothing(  
    ts_out, trend="add", seasonal="add", seasonal_periods=5  
)  
.fit()  
  
model_in = ExponentialSmoothing(  
    ts_in, trend="add", seasonal="add", seasonal_periods=5  
)  
.fit()  
  
model_out = ExponentialSmoothing(  
    ts_out, trend="add", seasonal="add", seasonal_periods=5  
)  
.fit()  
  
f_in = model_in.forecast(4)  
f_out = model_out.forecast(4)  
  
f_net = f_in + f_out
```

Code Snippet

f_net	
2025-11-03	640,900.92
2025-11-10	-1,319,783.68
2025-11-17	-1,313,788.60
2025-11-24	167,240.85
dtype: float64	

Weekly Net Cash Flow (Forecast 4 weeks)

f_in	
2025-11-03	3,148,174.07
2025-11-10	1,442,011.35
2025-11-17	3,410,451.75
2025-11-24	4,254,785.34
dtype: float64	

Weekly Inflow (Forecast 4 weeks)

f_out	
2025-11-03	-2,505,273.15
2025-11-10	-2,781,795.04
2025-11-17	-4,724,240.35
2025-11-24	-4,087,524.50
dtype: float64	

Weekly Outflow (Forecast 4 weeks)

Key Forecast Findings (Next 1 Month)

- Weekly inflows are expected to remain stable and seasonal, driven by regular billing cycles.
- Outflows continue to occur in large, predictable batches, reflecting scheduled payments.
- Net cash is forecasted to fluctuate around breakeven, with temporary deficits followed by recovery.
- No sustained negative cash trend is expected over the next four weeks

The one-month forecast indicates manageable short-term liquidity, with cash volatility driven by payment timing rather than declining inflows.

Six-Month Weekly Cash Flow Forecast & Liquidity Outlook

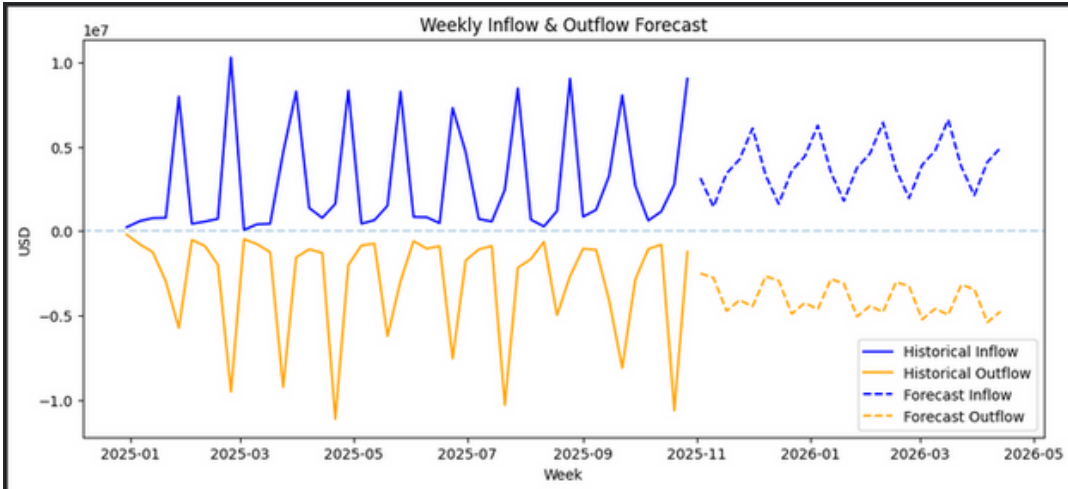


Chart: Weekly Inflow & Outflow Forecast

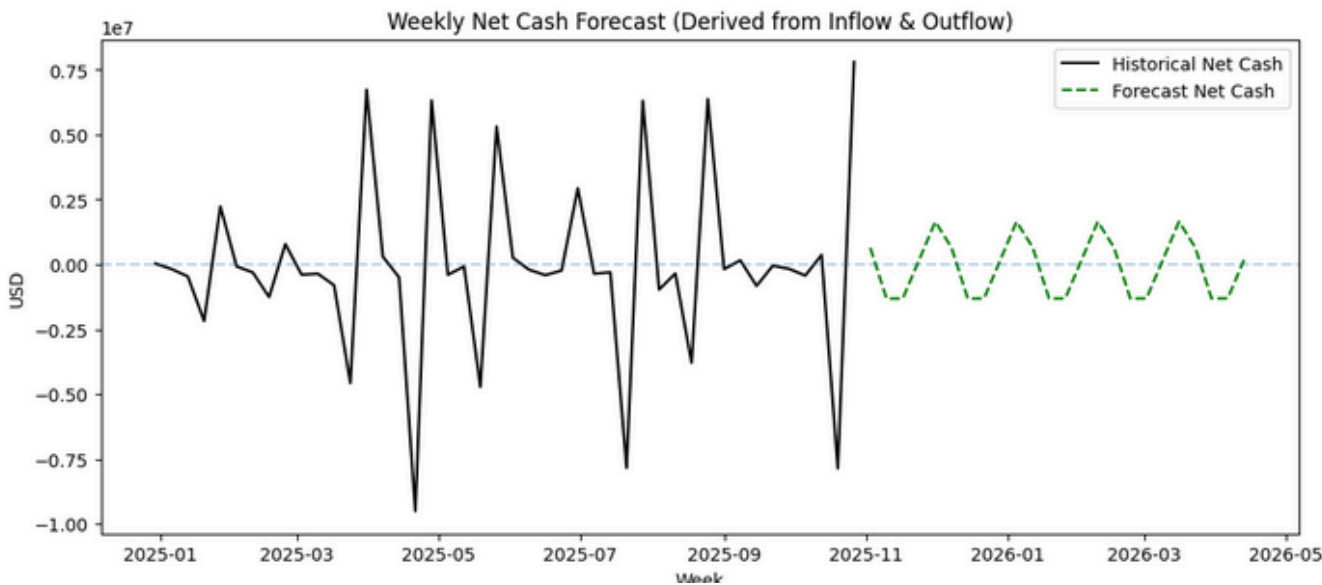


Chart: Weekly Net Cash Forecast (24 weeks)

```
model_in24 = ExponentialSmoothing(  
    ts_in, trend="add", seasonal="add", seasonal_periods=5  
) .fit()  
  
model_out24 = ExponentialSmoothing(  
    ts_out, trend="add", seasonal="add", seasonal_periods=5  
) .fit()  
  
model_in24 = ExponentialSmoothing(  
    ts_in, trend="add", seasonal="add", seasonal_periods=5  
) .fit()  
  
model_out24 = ExponentialSmoothing(  
    ts_out, trend="add", seasonal="add", seasonal_periods=5  
) .fit()  
  
f_in24 = model_in.forecast(24)  
f_out24 = model_out.forecast(24)  
  
f_net24 = f_in24 + f_out24
```

Code Snippet

f_net24	
2025-11-03	640,900.92
2025-11-10	-1,310,783.89
2025-11-17	-1,313,788.80
2025-11-24	167,240.85
2025-12-01	1,646,519.75
2025-12-08	642,457.80
2025-12-15	-1,318,228.81
2025-12-22	-1,312,231.72
2025-12-29	168,797.73
2026-01-05	1,648,070.64
2026-01-12	644,014.68
2026-01-19	-1,316,889.93
2026-01-26	-1,310,874.84
2026-02-02	170,354.61
2026-02-09	1,649,633.52
2026-02-16	645,571.56
2026-02-23	-1,315,113.05
2026-03-02	-1,309,117.98
2026-03-09	171,911.49
2026-03-16	1,651,190.40
2026-03-23	647,128.44
2026-03-30	-1,313,556.17

Weekly Net Cash Flow (Forecast 24 weeks)

f_in24	
2025-11-03	3,146,174.07
2025-11-10	1,442,011.35
2025-11-17	3,410,451.75
2025-11-24	4,254,765.34
2025-12-01	6,094,220.43
2025-12-08	3,315,945.45
2025-12-15	1,611,782.73
2025-12-22	3,580,223.14
2025-12-29	4,424,538.73
2026-01-05	6,284,000.82
2026-01-12	3,485,710.83
2026-01-19	1,781,554.11
2026-01-26	3,749,994.52
2026-02-02	4,594,308.11
2026-02-09	6,433,772.20
2026-02-16	3,655,488.21
2026-02-23	1,951,325.46
2026-03-02	3,919,765.90
2026-03-09	4,764,079.49
2026-03-16	6,603,543.58
2026-03-23	3,825,259.60
2026-03-30	2,121,090.88
2026-04-06	4,089,537.28

Weekly Inflow (Forecast 24 weeks)

f_out24	
2025-11-03	-2,505,273.15
2025-11-10	-2,781,795.04
2025-11-17	-4,724,240.35
2025-11-24	-4,087,524.50
2025-12-01	-4,447,709.68
2025-12-08	-2,673,487.65
2025-12-15	-2,930,009.54
2025-12-22	-4,892,454.85
2025-12-29	-4,255,739.00
2026-01-05	-4,615,924.18
2026-01-12	-2,841,702.15
2026-01-19	-3,098,224.04
2026-01-26	-5,060,669.39
2026-02-02	-4,423,953.50
2026-02-09	-4,784,138.68
2026-02-16	-3,009,916.65
2026-02-23	-3,268,438.54
2026-03-02	-5,228,883.88
2026-03-09	-4,592,168.00
2026-03-16	-4,952,353.18

Weekly Outflow (Forecast 4 weeks)

Key Findings (24 Weeks)

- Weekly inflows remain stable and strongly seasonal, indicating consistent customer collections.
- Weekly outflows continue to occur in large, predictable cycles, reflecting planned operational payments.
- There is no downward trend in net cash over the 24-week horizon.

The six-month forecast confirms a sustainable cash position, with predictable short-term deficits consistently offset by regular inflows.

Short-Term Cash Flow Outlook (Next 1 Month)

Short-Term Liquidity Focus

Expected Net Cash Position

- The one-month weekly forecast indicates that net cash will fluctuate around breakeven.
- Temporary negative weeks are expected, but these are short-lived and followed by recovery.
- No sustained cash deficit is forecasted over the next four weeks.

Primary Drivers of Short-Term Liquidity

- Large, scheduled outflows (e.g. operational and vendor payments) are the main source of short-term volatility.
- Inflows remain stable and predictable, driven by regular billing cycles.
- Liquidity risk is therefore driven by payment timing, not revenue weakness.

Long-Term Cash Flow Outlook (Next 6 Months)

Cash Position Sustainability

- The six-month weekly forecast shows a stable and repeating seasonal pattern.
- Net cash alternates between surplus and deficit weeks, with no deterioration trend.
- Forecast accuracy against historical data supports confidence in the medium-term outlook.

Structural Cash Flow Patterns to Monitor

- Recurring payment cycles that create predictable negative cash weeks.
- Alignment between inflow timing and outflow schedules, especially during peak payment periods.
- Concentration of large payments that may temporarily strain liquidity.